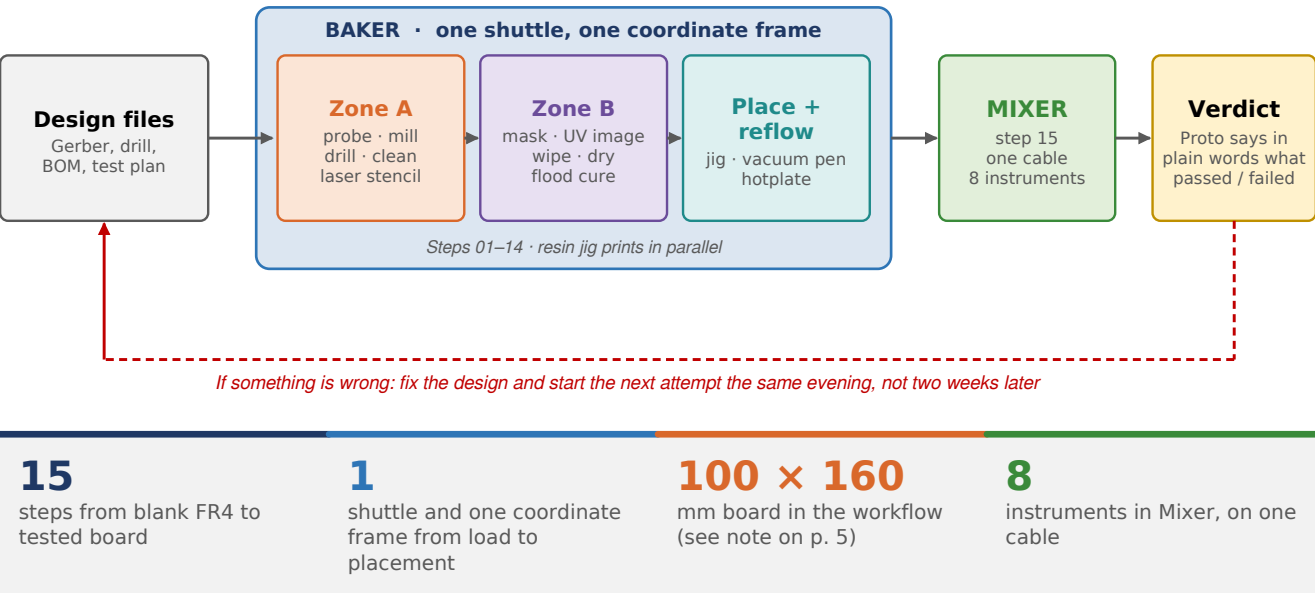


The story of a PCB

From a blank sheet of copper-clad FR4 to an assembled, tested board, and a plain-language verdict, on two machines on one bench.



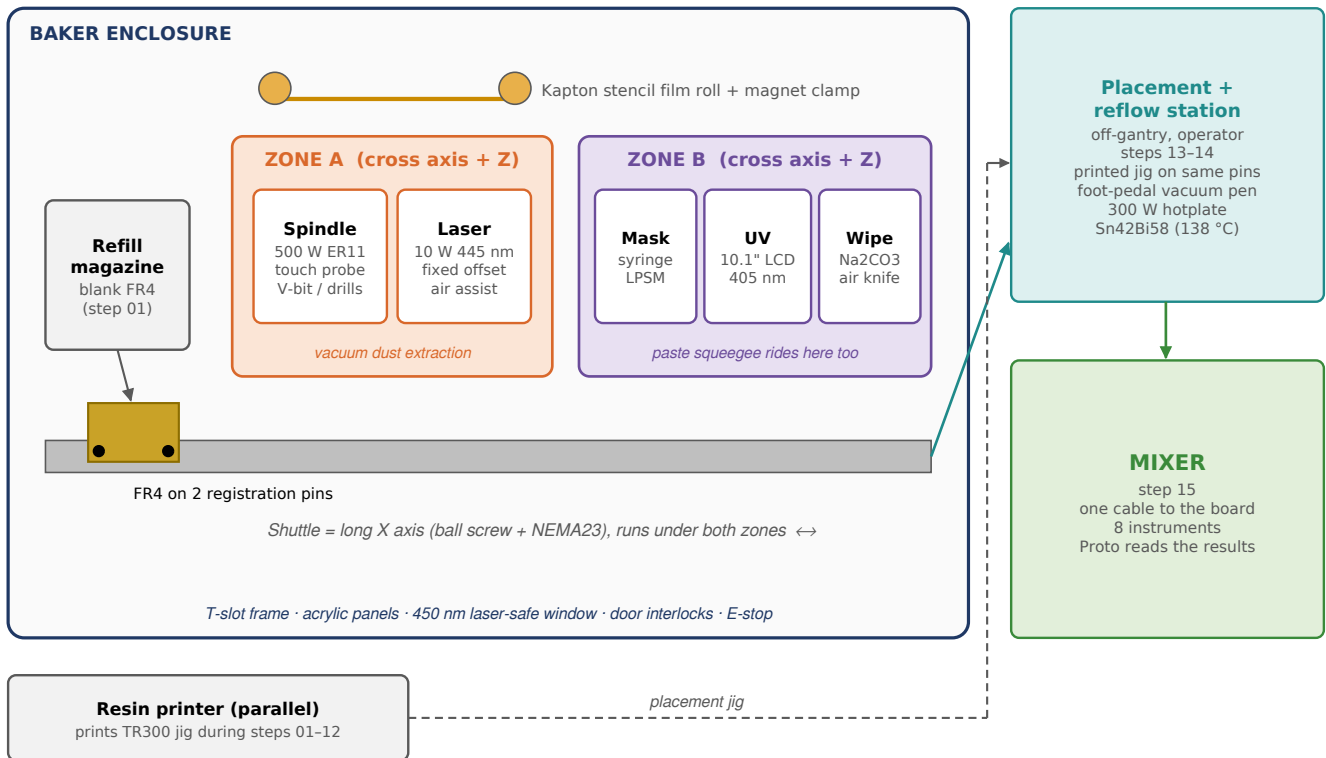
Baker makes the board. Zone A (spindle and laser) cuts copper, drills and cuts a stencil; Zone B (dispenser, UV screen and wiper) turns bare copper into a solder-masked surface; the operator places parts into a printed jig and the hotplate reflows them. **Mixer** then proves the board works: it powers it, drives it, measures it and tells you in plain words what passed and what did not.

This document tells that story twice: first as a map (what happens where), then as a narrative that follows one board through every step, with the hardware that does each job, the reasons behind the order, and the points that still need to be proven on the prototype.

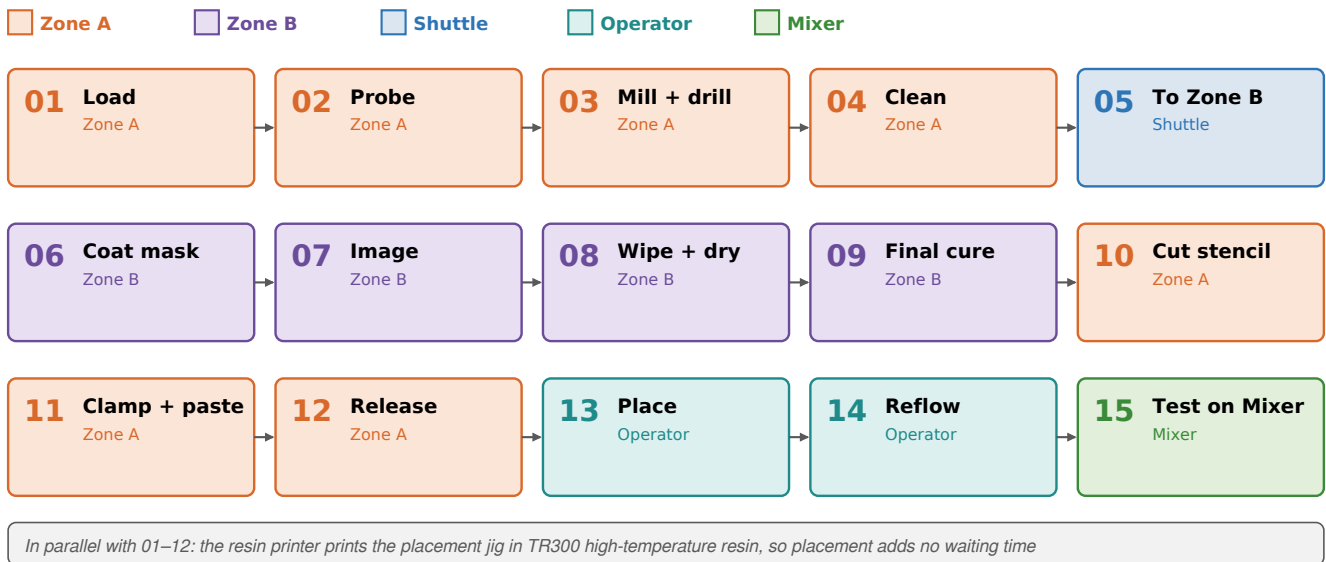
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1 Baker at a glance

Fifteen steps on one shuttle, in one box, then onto Mixer. The shuttle is Baker's long X axis: it runs under both zones, so the board never leaves its fixture until the parts go on. Everything from step 01 to step 12 happens in a single coordinate frame, set by two registration pins.



The 15 steps



2 The story of one board on Baker

Follow a single 100 × 160 mm board from the magazine to the hotplate. Each chapter lists the “cast”: the parts from the Baker BOM that do the work.

1 It starts as a plain sheet

Shuttle · steps 01, 05

A copper-clad board drops out of the refill magazine onto the shuttle and seats on two registration pins. From here until placement, every step shares one coordinate frame, which is what makes the later steps line up.

The shuttle is the long X axis. It carries the board under Zone A for cutting, under Zone B for masking, and back under Zone A for the stencil, without the board ever being lifted or re-registered.

Cast

- FR4 copper clad blanks
- Registration dowel pins and springs
- Shuttle: ball-screw stage + NEMA23
- 6 mm aluminium shuttle fixture plate
- Endstops for homing

2 Zone A reads the board, then cuts it

Zone A · steps 02–04

FR4 is never perfectly flat, so the spindle’s touch probe taps a grid of points and builds a height map first. The isolation path then follows that map, cutting through the copper by the same few microns everywhere.

The same spindle drills the holes. A vacuum pulls the dust away so none of it reaches the mask.

Cast

- Z touch probe (pogo pin + clip)
- 500 W ER11 spindle, collet set
- 20° 0.1 mm V-bits (isolation)
- Carbide drills 0.3–1.2 mm
- 1000 W wet/dry vacuum
- Cross axis + SFU1605 Z stage

Why probe first: without a height map, a 0.1 mm V-bit cuts too deep where the board bows up and misses copper where it bows down.

3 Zone B turns copper into a finished surface

Zone B · steps 06–09

The shuttle carries the board under the dispenser, which lays a thin coat of alkali-developing solder mask over everything.

The 10.1-inch LCD above it shows an inverse image of the pads, so UV cures all the mask except the pad openings. A dilute sodium-carbonate wipe lifts the soft mask off the pads, an air knife dries the board, and a flood exposure hardens the edges.

No tank, no bath: the only liquid is a few millilitres on the squeegee.

Cast

- 30 cc syringes + blunt needles
- Alkali-developing LPSM
- 10.1" 12K mono LCD + ChiTu L K1 board
- 405 nm parallel UV light board
- 12 V peristaltic pump (developer)
- Squeegee blade, 5015 blowers (air knife)
- Waste tray

4 The stencil is made to fit this exact board

Zone A · steps 10–12

Back under Zone A, the laser, fixed at a known offset beside the spindle, cuts the pad openings into the film running just above the board.

The film advances into place, aligns, and four electromagnets pull its steel frame flat against the board. The squeegee drags paste across, the film lifts clean, a solenoid blade cuts off the used piece, and the roll is ready for the next board.

Why cut the stencil in place: it is cut from the same Gerber, in the same frame as the milling, which removes the usual stencil-to-board alignment error.

Cast

- 10 W 445 nm diode laser + air assist
- Polyimide (Kapton) film roll
- 4 × 12 V 8 kg holding electromagnets
- Ferrous 430 clamp frame
- 25 N push-pull solenoid (cut-off)
- Sn42Bi58 low-temp paste

5 Placement is fast because the jig does the aiming

Operator · steps 13

While all of this ran, the resin printer printed a placement jig in high-temperature resin: a plate with one pocket for every part, located by the same pins.

The operator puts the jig on, picks each part with the foot-pedal vacuum pen and drops it into its pocket. Paste tack holds the parts.

Cast

- Resin printer + TR300 resin (HDT 160 °C)
- 12 V vacuum pump + NC solenoid valve
- Foot pedal switch
- Manual vacuum pen (backup)

6 Reflow at low temperature

Operator · steps 14

The jig lifts off, and the hotplate runs a low-temperature reflow profile. Low-temperature Sn42Bi58 paste (melts at 138 °C) keeps the hotplate below the resin's 160 °C limit when the jig is left nearby, and reduces board warp.

Then the board goes to Mixer: it gets tested, not just made.

Cast

- 300 W 230 V silicone heater pad
- SSR-25DA solid-state relay
- MAX31855 + K-type thermocouple
- 5015 blowers (cool-down)

3 Why this order, and what is still open

Why this order

Rule	Reason
Milling comes before masking	The mask covers finished, clean copper.
The mask is cured before the stencil is cut	The laser never touches uncured mask.
The stencil is cut from the same Gerber in the same frame as the milling	Removes the usual stencil-alignment error.
The jig prints in parallel (steps 01–12)	Placement adds no waiting time.
Low-temperature paste (Sn42Bi58, 138 °C)	Keeps the hotplate below the resin's 160 °C limit when the jig is nearby, and reduces board warp.
Jig lifts off before reflow	TR300 is rated to 160 °C, below reflow; paste tack holds the parts. A jig that rides through reflow would need resin rated above 200 °C, which costs several times more.

Open points to prove in the prototype, written as acceptance tests

The workflow names four open points. Each one is written below as a pass/fail test, so the prototype result can go straight into the production specification and into the factory's acceptance plan.

ID	Open point	Proposed test	Pass criterion	Fallback
PR-001	Wipe-develop cleanliness on 0.5 mm-pitch pads	Run a test board with a 0.5 mm-pitch footprint; inspect every pad under magnification after step 09	No residual mask on any defined pad area	Skim the pads with the spindle
PR-002	Film registration after magnetic clamping	Cut fiducial openings; measure film-to-copper offset at the fiducials after clamping	Within $\pm 50 \mu\text{m}$ at every fiducial	Adjust clamp sequence / frame stiffness
PR-003	Jig-to-pin accuracy after resin shrink	Print jig, fully post-cure, measure pocket positions against pin datum	Pocket error small enough that parts land on pads (limit to set on prototype)	Scale compensation in the jig model
PR-004	Laser offset drift with temperature	Re-check laser-to-spindle offset against a fiducial at cold start and after a long job	Drift inside the stencil budget; check runs every job	Fiducial re-check every job (already planned)

One inconsistency to resolve

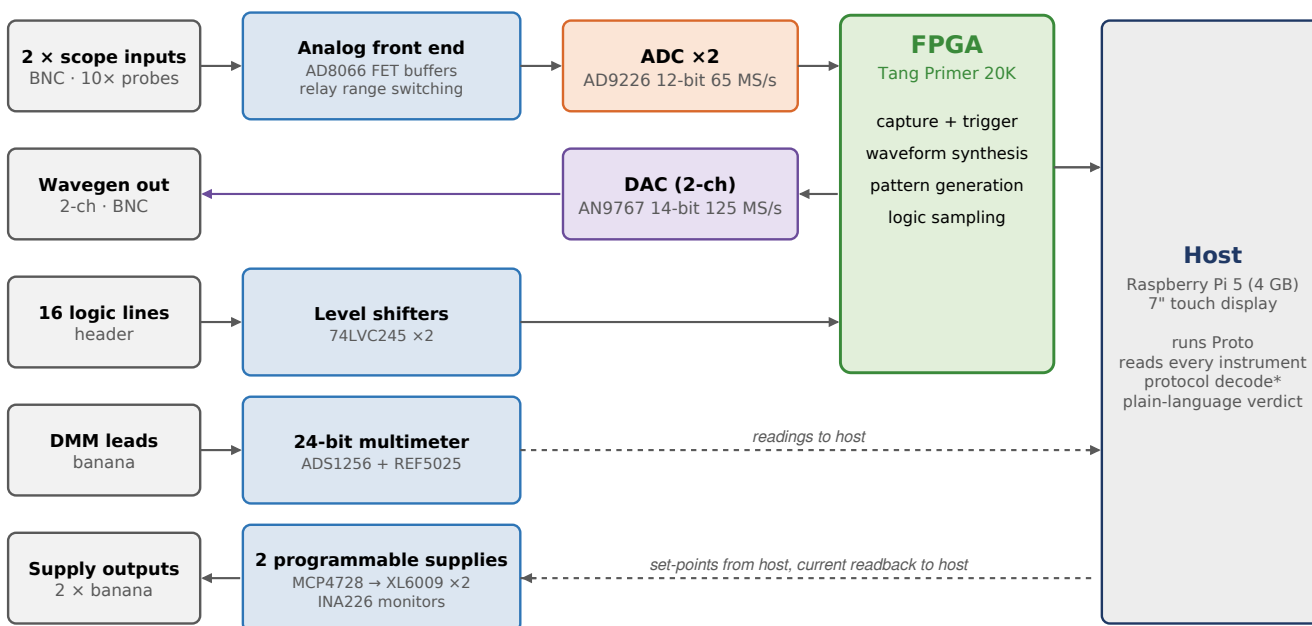
The workflow story says **100 × 160 mm** boards; the BOM buys **100 × 150 mm** FR4 blanks. Pick one before the magazine, fixture plate and jig are frozen for production.

4 Mixer at a glance

Mixer is a lower-cost instrument in the Analog Discovery 3 class: eight instruments on one host, connected to the board by one cable. On Baker's bench it is step 15, the moment a made board becomes a proven board.

Instrument	Prototype specification	Key parts (prototype BOM)
1 Oscilloscope	2 channels, 12-bit, 65 MS/s	AD9226 ADC ×2, AD8066 buffers, relays, BNC, 100 MHz probes
2 Waveform generator	2 channels, 14-bit	AN9767 dual 125 MS/s DAC
3 Logic analyser	16 channels	74LVC245 level shifters ×2 into the FPGA
4 Pattern generator	Digital stimulus	FPGA + level shifters
5 Protocol decoder	Reads captured bus traffic	FPGA / host software
6 Programmable supply 1	Set-point + current readback	MCP4728 DAC, XL6009 buck-boost, INA226
7 Programmable supply 2	Set-point + current readback	MCP4728 DAC, XL6009 buck-boost, INA226
8 Multimeter	24-bit	ADS1256 + REF5025 precision reference
Host	Runs Proto, reads every instrument	Raspberry Pi 5 4 GB, 7" Touch Display 2, Tang Primer 20K FPGA

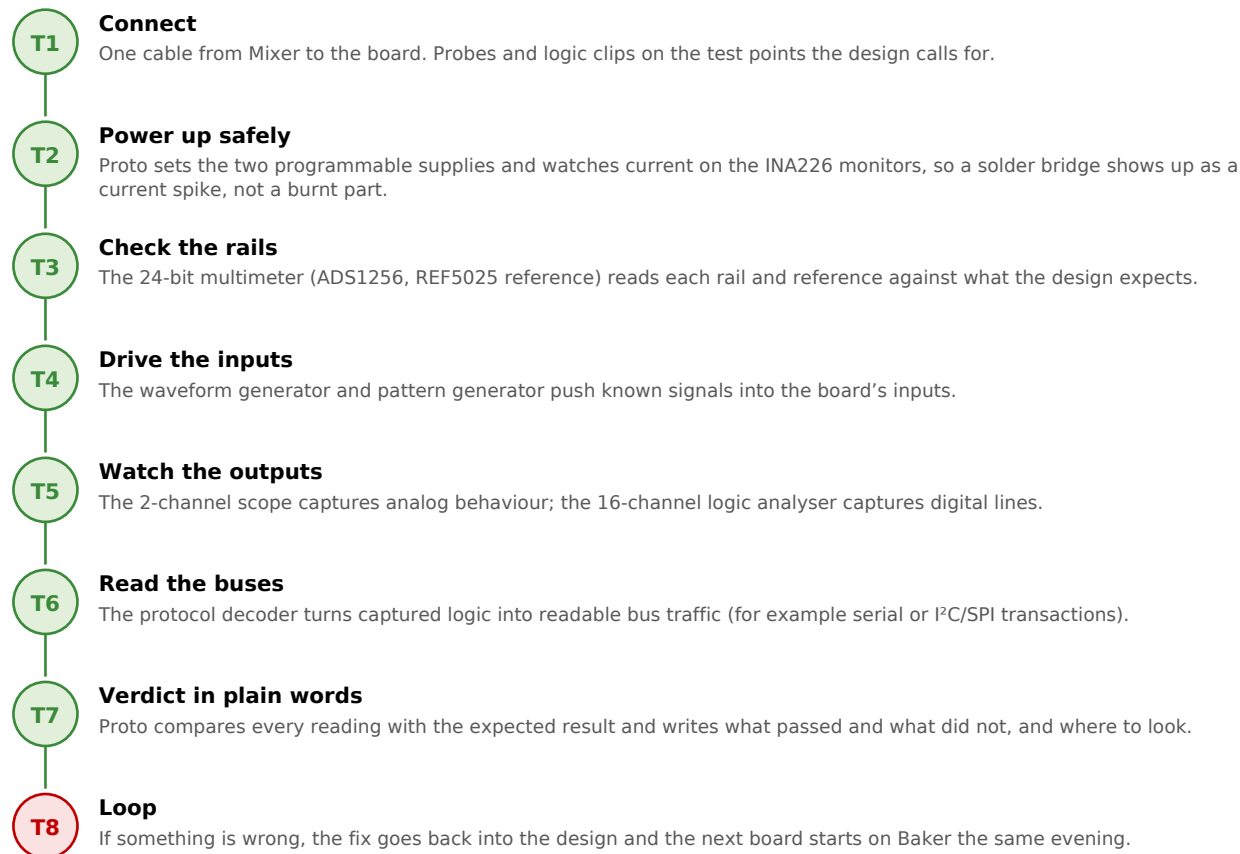
Signal path inside Mixer



Prototype signal path, drawn from the Mixer BOM. *Protocol decoding is listed as an instrument; whether it runs in the FPGA or on the host is a firmware choice.

5 The story continues on Mixer

The board from chapter 2 has just come off the hotplate. It is assembled, but nobody knows yet whether it works. Mixer answers that. Proto sets the supplies, drives the inputs, reads the scope and logic channels, and says in plain words what passed and what did not.



Steps T1–T8 describe how Mixer's instruments are used together, in the order a careful engineer would use them. The exact checks for each board (which rails, which signals, which limits) come from that board's own design and test plan.

One evening, one loop

The point of putting Baker and Mixer on one bench is the loop time. A normal prototype cycle sends Gerbers out, waits for boards, assembles them, then borrows lab instruments to debug. Here the same person goes from design files to a tested board to a plain-language verdict, and when something is wrong, **the next attempt starts the same evening, not two weeks later.**

	Usual prototype loop	Baker + Mixer loop
Board fabrication	Outside fab, shipped	Baker steps 01–12, on the bench
Stencil	Ordered separately, aligned by hand	Cut in place from the same Gerber, same frame
Placement	Tweezers or a pick-and-place service	Printed jig + vacuum pen
Test	Separate scope, supply, DMM, logic analyser	Mixer: one cable, eight instruments
Diagnosis	Engineer interprets raw traces	Proto states what passed and failed in plain words
Next attempt	About two weeks later	The same evening

6 Prototype → production

The workflow above is the prototype. For production, both machines keep the same story but change the parts that tell it, mainly to reach the pitch-deck hardware targets. Costs are from the Protopus Cost Report (29 Sep 2026).

Baker

Function	Prototype	Production	Why
Axes	Finished ball-screw linear stages	SFU1605 ball screws + HGR15 rails built in	About ₹5,500 per axis instead of ₹16,500
Control	Raspberry Pi 5 + BTT Octopus Pro + DM556 drivers	BTT Manta M8P V2 + CB1 compute module	One board with 8 drivers, about ₹10,800
Power	24 V + 48 V + 12 V supplies	One 24 V rail (spindle keeps its own)	Fewer supplies, simpler wiring
Placement jig	Separate Elegoo Saturn 3 Ultra printer + wash/cure	Small resin vat + lift under Baker's own 10.1" UV screen	Removes the separate printer (flagged for testing)
Frame	T-slot extrusion + acrylic	Sheet-metal frame + acrylic panels	Repeatable fabrication
Unit cost	₹3,73,418 (with 15% contingency)	₹1,44,842 (China parts, Delhi assembly)	Deck target ₹1,50,000

Mixer

Function	Prototype	Production	Why
Capture	AD9226 12-bit 65 MS/s ×2	AD9288 dual 8-bit 80 MS/s	Cost; prototype keeps 12-bit so the front end can be tuned first
Generation	AN9767 dual 14-bit	DAC908 8-bit 165 MS/s ×2	Cost
FPGA	Tang Primer 20K Dock	Tang Nano 9K (GW1NR-9 + 64 Mb PSRAM)	Smaller, cheaper module
Host	Raspberry Pi 5 + 7" screen on board	FT232HL USB 2.0 high-speed; no screen of its own	One USB cable to the user's computer
Enclosure	Vacuum-formed HIPS shell	Extruded aluminium case	Production finish
Unit cost	₹74,847 (with 15% contingency)	₹13,740 (China parts, Delhi assembly)	Deck target ₹15,000

Mixer as the production tester

Because Mixer already measures supplies, analog and digital signals and reports pass/fail, the same Mixer and Proto software can become the **production tester** for boards and machines built by a contract manufacturer. The China assembly-line plan builds on this.

Sources: Protopus Baker Workflow Story (29 Sep 2026), Protopus Cost Report (29 Sep 2026), Protopus BOM Baker + Mixer workbook.